Annex III. Public Sector Debt Sustainability Analysis[1]

1. Summary: Public debt sustainability risks remain contained due to the low stock of public debt and conservative spending, underpinned by conservative implementation of fiscal rules. Switzerland has substantial fiscal space, both in economic terms and within the debt brake rule. Public finances have run small structural surpluses leading to declining general government gross debt. However, prospective aging of the population will weigh on Switzerland's future potential output and fiscal situation. For most indicators on fiscal space Switzerland remains below benchmark levels, except for long-term adjustment needs to address the cost of aging. Switzerland's debt reduction has been anchored on its fiscal rule.

2. Baseline scenario: The key assumptions underlying the baseline scenario are a gradual but steady recovery of economic growth and the continued adherence to federal and sub-federal fiscal rules. Under the baseline scenario, public debt is projected to decline from 41.9 percent of GDP in 2016 to below 34 percent of GDP in 2023. Gross financing needs are expected to decline from 2.6 percent of GDP in 2016 to around 1.7 percent of GDP during the medium term.

3. Stress tests: The main risk to debt dynamics is a negative growth shock. Other risks such as an adverse interest rate shock or a shock to financing needs affect the public debt trajectory only to a minor extent.

- **Real GDP growth shock.** Real GDP growth rates are assumed to be one standard deviation (1.4 percent) below the baseline during 2019–20. Under this scenario, the debt-to-GDP ratio rises to 40.4 percent in 2020 (about 3 percentage points higher than the baseline).
- **Primary balance shock.** The primary balance in 2019–20 is hit by a negative shock of 0.4 percent of GDP. This shock results in a debt-to-GDP ratio that is about 0.8 percentage point above the baseline during 2020–23.
- **Real interest rate shock**. The nominal interest rate increases by 200 basis points during 2019–23. The debt-to-GDP ratio becomes slightly higher than the baseline but continues declining.
- **Real exchange rate shock**. This scenario assumes that the nominal CHF/USD exchange rate increases by 9 percent in 2019 relative to its 2016 level. As with other non-growth-related shocks, the impact of this shock on the trajectory of public debt is minor.
- **Combined shock**. A simultaneous combination of the previous three shocks would result in debt-to-GDP ratio that approached 40.4 percent in 2020 (approximately 3 percentage points higher than the baseline). However, after 2020, the debt would start declining on a trajectory that is parallel to the one under the baseline scenario.

[1] General government includes the federal government, cantonal governments, municipal governments, and the social security fund.

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Figure 1. Switzerland Public Sector Debt Sustainability Analysis (DSA)—Baseline Scenario

(In percent of GDP, unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

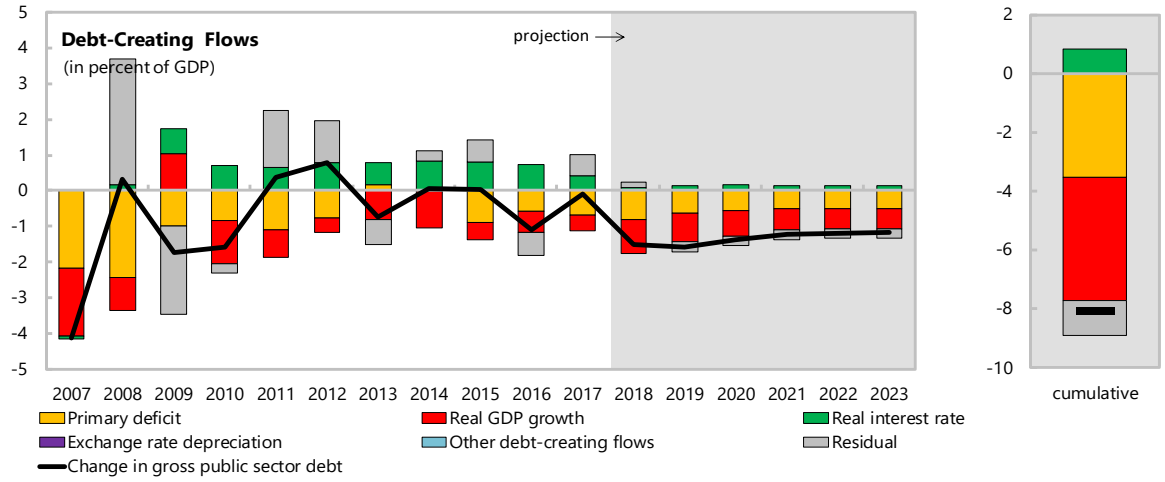
	Actual			Projections					
	2007-2015 [2/]	2016	2017	2018	2019	2020	2021	2022	2023
Nominal gross public debt	43.7	41.9	41.8	40.3	38.7	37.3	36.1	34.9	33.8
Public gross financing needs	1.8	2.6	2.6	2.2	2.2	1.9	1.8	1.7	1.7
Real GDP growth (in percent)	1.7	1.4	1.1	2.3	2.0	1.9	1.7	1.6	1.7
Inflation (GDP deflator, in percent)	0.5	-0.6	0.3	1.1	1.1	1.0	1.0	1.0	1.0
Nominal GDP growth (in percent)	2.2	0.8	1.4	3.5	3.1	2.9	2.7	2.7	2.7
Effective interest rate (in percent) [4/]	1.8	1.1	1.4	1.4	1.5	1.4	1.4	1.4	1.5

As of May 02, 2018

Sovereign Spreads		
EMBIG (bp) 3/		-44
5Y CDS (bp)		11
Ratings	Foreign	Local
Moody's	Aaa	Aaa
S&Ps	AAA	AAA
Fitch	AAA	AAA

Contribution to Changes in Public Debt

	Actual			Projections							
	2007-2015	2016	2017	2018	2019	2020	2021	2022	2023	cumulative	debt-stabilizing primary balance
Change in gross public sector debt	-0.7	-1.1	-0.1	-1.5	-1.6	-1.4	-1.2	-1.2	-1.2	-8.1	
Identified debt-creating flows	-1.2	-0.4	-0.7	-1.7	-1.3	-1.1	-1.0	-0.9	-0.9	-6.9	
Primary deficit	-1.0	-0.6	-0.7	-0.8	-0.6	-0.6	-0.5	-0.5	-0.5	-3.5	-0.4
Primary (noninterest) revenue and gra	32.1	33.1	33.1	33.1	33.1	33.1	33.1	33.1	33.2	198.8	
Primary (noninterest) expenditure	31.1	32.5	32.4	32.3	32.5	32.6	32.6	32.6	32.6	195.3	
Automatic debt dynamics [5/]	-0.2	0.1	0.0	-0.8	-0.7	-0.6	-0.5	-0.4	-0.4	-3.4	
Interest rate/growth differential [6/]	-0.2	0.1	0.0	-0.8	-0.7	-0.6	-0.5	-0.4	-0.4	-3.4	
Of which: real interest rate	0.6	0.7	0.4	0.1	0.1	0.2	0.1	0.1	0.2	0.8	
Of which: real GDP growth	-0.7	-0.6	-0.4	-0.9	-0.8	-0.7	-0.6	-0.6	-0.6	-4.2	
Exchange rate depreciation [7/]	0.0	0.0	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
0 (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Please specify (2) (e.g., ESM and Euro	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes [8/]	0.4	-0.7	0.6	0.1	-0.3	-0.3	-0.3	-0.3	-0.3	-1.2	



Source: IMF staff.

1/ Public sector is defined as general government.

2/ Based on available data.

3/ Long-term bond spread over German bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

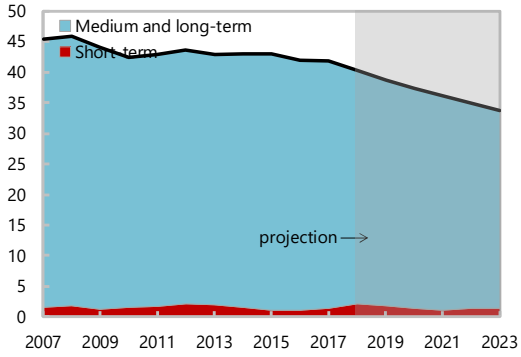
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Figure 2. Switzerland Public DSA—Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

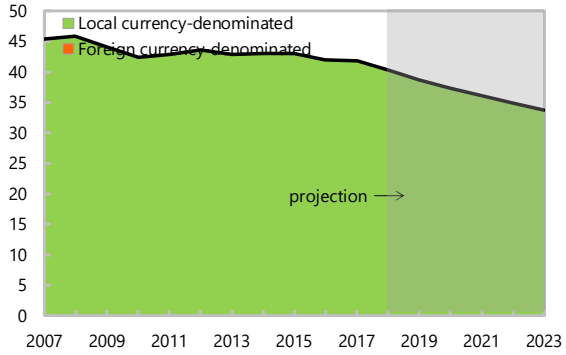
By Maturity

(in percent of GDP)

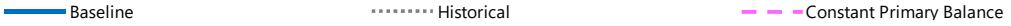


By Currency

(in percent of GDP)

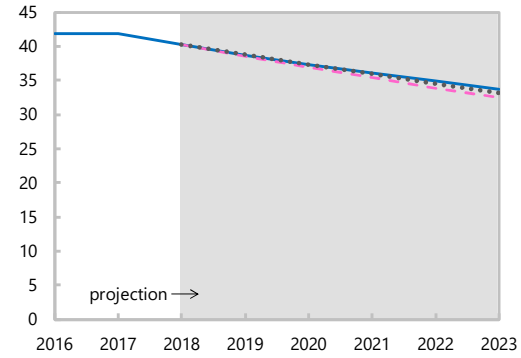


Alternative Scenarios



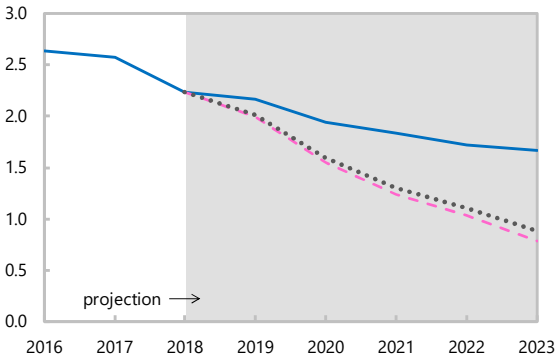
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2018	2019	2020	2021	2022	2023
Real GDP growth	2.3	2.0	1.9	1.7	1.6	1.7
Inflation	1.1	1.1	1.0	1.0	1.0	1.0
Primary Balance	0.8	0.6	0.6	0.5	0.5	0.5
Effective interest rate	1.4	1.5	1.4	1.4	1.4	1.5
Constant Primary Balance Scenario						
Real GDP growth	2.3	2.0	1.9	1.7	1.6	1.7
Inflation	1.1	1.1	1.0	1.0	1.0	1.0
Primary Balance	0.8	0.8	0.8	0.8	0.8	0.8
Effective interest rate	1.4	1.5	1.4	1.4	1.4	1.5

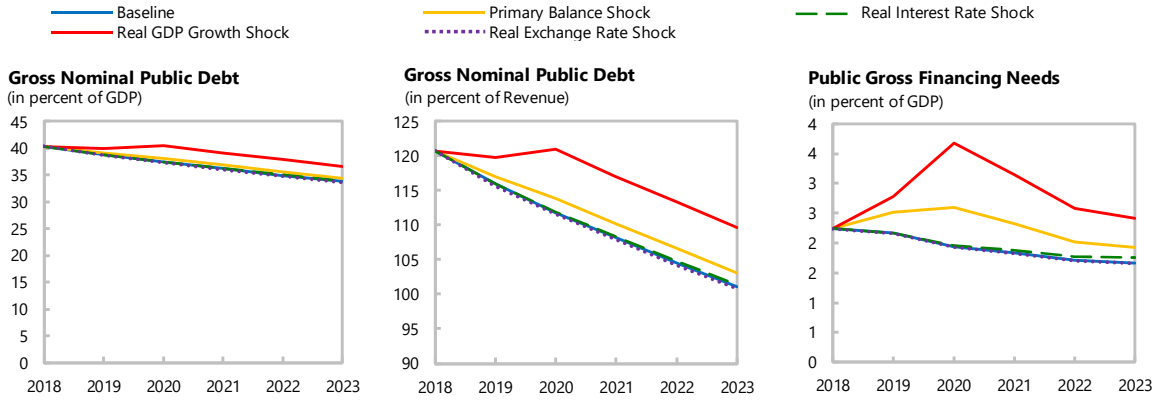
Historical Scenario	2018	2019	2020	2021	2022	2023
Real GDP growth	2.3	1.4	1.4	1.4	1.4	1.4
Inflation	1.1	1.1	1.0	1.0	1.0	1.0
Primary Balance	0.8	0.8	0.8	0.8	0.8	0.8
Effective interest rate	1.4	1.5	1.5	1.5	1.5	1.5

Source: IMF staff.

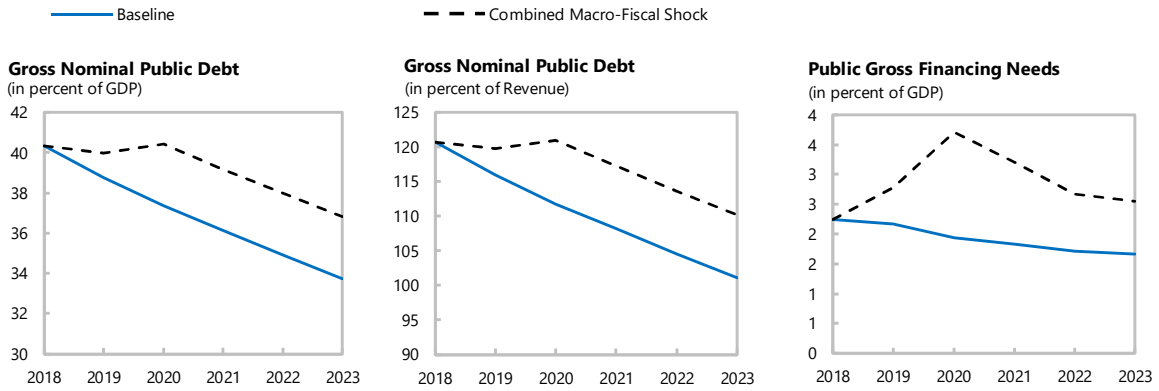
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Figure 3. Switzerland Public DSA—Stress Tests

Macro-Fiscal Stress Tests



Additional Stress Tests



Underlying Assumptions

(in percent)

Primary Balance Shock	2018	2019	2020	2021	2022	2023
Real GDP growth	2.3	2.0	1.9	1.7	1.6	1.7
Inflation	1.1	1.1	1.0	1.0	1.0	1.0
Primary balance	0.8	0.3	0.2	0.5	0.5	0.5
Effective interest rate	1.3	1.4	1.4	1.4	1.4	1.4
Real Interest Rate Shock						
Real GDP growth	2.3	2.0	1.9	1.7	1.6	1.7

Real GDP Growth Shock	2018	2019	2020	2021	2022	2023
Real GDP growth	2.3	0.6	0.5	1.7	1.6	1
Inflation	1.1	0.8	0.7	1.0	1.0	1
Primary balance	0.8	0.1	-0.6	0.5	0.5	0
Effective interest rate	1.3	1.4	1.4	1.4	1.3	1
Real Exchange Rate Shock						
Real GDP growth	2.3	2.0	1.9	1.7	1.6	1

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